Alex Dashboard Edits

Comprehensive Test and Verification Report

Groups 1–9, configuration tiering, exports, modeling engine, and time-period analysis

Final assessment: All items from Alex's annotated PowerPoint have passed the final verification state documented in this report.

Current dashboard wiring: 66 components / 27 callbacks.
Current sponsor-estimator check: PASS; coefficient gap approximately 1.7×10^{-8} of the largest coefficient scale.

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Report date:	July 23, 2026
Repository:	`C:\Users\BoqiNiu\Claude\Projects\Capstone`
Primary dataset:	`AnonymizedDataforProject.xlsx`

Contents

1 Executive Summary

This report consolidates the full test history from the implementation and review session for Alex's dashboard feedback. It records the initial positive-path checks, every material defect uncovered during review, the corresponding corrective change, and the successful retest. The catalog contains **184 named test cases**; many cases contain multiple assertions.

Final result. The dashboard changes in Groups 1–9, including the two-tier variable configuration model and CLI unification, passed the final state described here. The final current-state smoke test produced:

- Dashboard registration: **66 components / 27 callbacks**.
- Python compilation: all five modified source files compiled successfully.
- Sponsor estimator equivalence: **PASS**, with a scale-normalized coefficient gap of approximately 1.69×10^{-8} in the final rerun.
- Model-window regression: 52-, 104-, and 156-week runs completed successfully.
- Three-year robustness: the 156-week run produced three model-year periods.
- Configuration hygiene: no config diffs and no upload temp, backup, or test override artifacts remained after verification.

Important scope note. The runtime/scaling benchmark requested verbally in the July meeting (1 model, 10 models, full set, and doubled run) is a separate deliverable and is not claimed as completed by this report. The screenshots had already been sent to Alex and are also outside this report's test scope.

2 Evidence Model and Confidence

Each test is tagged with one or more evidence classes:

Tag	Evidence class	Meaning
I	Implementation evidence	A test reported during implementation, including targeted fault injection, temporary backup/restore, callback inspection, or behavioral assertions.
R	Independent review evidence	A test independently repeated during this review conversation, generally read-only against the working tree and current dataset.
F	Final regression evidence	A current-state rerun after all fixes, including compilation, callback registration, selected behavioral checks, equivalence, and workspace hygiene.

The report deliberately distinguishes reported implementation tests from independently reproduced tests. A test marked only **I**was not falsely represented as independently rerun. Tests marked **R**or **F**were directly reproduced during the review.

3 System Under Test

3.1 Modified source and tracking artifacts

- `code/dashboard.py`

- `code/capstone_pipeline.py`
- `code/run_all.py`
- `code/run_brand1_channel1.py`
- `code/reference_alex_curvefit.py`
- `docs/Alex_Dashboard_Edits_Plan.md` (tracker document, not a source file)

3.2 Final regression environment

Component	Version/value	Notes
Operating system	Windows	Local project workspace
Python	3.13.14	Final regression interpreter
NumPy	2.5.1	Numerical arrays
pandas	3.0.3	Workbook/config processing
SciPy	1.18.0	Constrained model fitting
Dash	4.4.0	Dashboard callbacks and layout
Plotly	6.8.0	Figures and chart traces
Dataset size	13,530,148 bytes	`AnonymizedDataforProject.xlsx`
Dataset SHA-256	`C4303D04...D7B31A0`	Full hash retained in the session evidence

4 Requirement Traceability

Group	Requirement area	Primary verification	Final state
1	Run-model restructure	Button placement, tab redirect, Model Runs rename, callback registration	**PASS**
2	Batch grade labels and window help	Threshold labels, cell fills, no-holdout state, tooltip content	**PASS**
3	k/M formatting and chart labels	Formatter boundary cases and both contribution figures	**PASS**
4	Diagnostics VIF	Final selected design matrix, forced variables, high/infinite display	**PASS**
5	Variables context	Coefficient/due-to context, period toggle, slice gating, edit preservation	**PASS**
5.5	Two-tier config	Default/override precedence, scoped writes, races, CLI parity, shared lock	**PASS**
6	Variable template and Export tab	Round-trip, validation, transaction rollback, three-sheet workbook	**PASS**
7	Definitions and Methods	Code-aligned formulas, thresholds, method wording, navigation	**PASS**
8	Time selection	Built-in periods, mapping upload, stale safety, period comparison, A-vs-A guard	**PASS**

Group	Requirement area	Primary verification	Final state
9	Final full-window model	Reserved validation tail, same selected structure, full-window refit, 3-year run	**PASS**

5 Detailed Test Catalog

5.1 A. Requirements, Baseline, and Structural UI

ID	Objective	Method / scenario	Result and evidence
`REQ-01`	Extract all requested edits	Compared the six-slide annotated PowerPoint against the living tracker and code areas.	All callouts assigned to Groups 1–9. **PASS I**
`REQ-02`	Resolve superseded requirements	Confirmed Alex's coefficient plus due-to specification superseded the temporary "was to now" design.	Temporary comparison removed from final scope. **PASS I**
`REQ-03`	Preserve source distinction	Separated PowerPoint requirements, July meeting statements, and internal team decisions.	VIF warning source and grade-threshold ownership recorded accurately. **PASS I**
`REQ-04`	Check tracker completeness	Cross-checked every tab section: Modeling, Diagnostics, Variables, Contributions, Batch, Extras, Export.	No PowerPoint callout left untracked. **PASS I**
`REQ-05`	Check final tracker state	Reviewed final Group 8 entries after E3/E4 completion.	C1, E3, E4 and Group 8 marked complete. **PASS R**
`REQ-06`	Protect unrelated workspace data	Repeatedly inspected Git/config state after tests and avoided reverting the pre-existing batch summary change.	No unintended source/config reversion. **PASS I R**
`UI-01`	Remove global Run button	Inspected layout placement after restructure.	Diagnostics and Contributions no longer expose a global Run command. **PASS I**
`UI-02`	Keep Run in authorized screens	Checked Variables and Model Runs layouts.	Run is present only where tuning or batch execution is intended. **PASS I**
`UI-03`	Place batch controls together	Checked Model Runs action layout.	Run model and Run all combinations are adjacent. **PASS I**

ID	Objective	Method / scenario	Result and evidence
`UI-04`	Redirect after Run	Triggered the store-driven tab transition for the same slice.	Run redirects to Diagnostics without changing slice context. **PASS I**
`UI-05`	Rename Batch navigation	Inspected the left navigation label.	Navigation displays Model Runs. **PASS I**
`UI-06`	Validate tab renderer	Registered all screens and switched tabs through the central store-driven renderer.	No duplicate or missing screen renderer. **PASS I**
`UI-07`	Register app wiring after Group 1	Imported the app and validated IDs/callback targets.	Initial restructure registered cleanly. **PASS I**
`UI-08`	Model grade labels	Exercised holdout values at good, moderate, and bad boundaries.	Good, Moderate, and Bad labels matched configured thresholds. **PASS I**
`UI-09`	No-holdout label	Passed a missing/NaN holdout metric to the grade helper.	Neutral No holdout state returned. **PASS I**
`UI-10`	Grade cell fills	Inspected Dash conditional styles for green, amber, red, and neutral states.	Whole grade cell receives the intended fill and readable text. **PASS I**
`UI-11`	Window tooltip	Rendered the WINDOW picker help after model-method revisions.	Tooltip explains model window, reserved validation tail, and due-to periods. **PASS I**
`UI-12`	Shared k/M formatter	Checked representative values 1,200,000; 550,000; and 10,675.	Outputs followed 1.2m, 550k, and 10.68k style. **PASS I**
`UI-13`	Totals chart labels	Inspected generated Plotly bars.	Due-to totals chart includes formatted data labels. **PASS I**
`UI-14`	Average chart labels	Inspected generated Plotly bars.	Average weekly due-to chart includes formatted data labels. **PASS I**
`UI-15`	Current final callback graph	Ran `_validate_wiring()` after every final fix.	Final result: 66 components / 27 callbacks. **PASS F**

5.2 B. Diagnostics and Variables Context

ID	Objective	Method / scenario	Result and evidence
VAR-01	Expose final-model VIF	Traced `fit.vif` to the final `X[selected]` design matrix.	VIF values correspond to variables actually in the model. **PASS I**
VAR-02	Include forced variables in VIF	Traced forced variables through selection seeding and no-prune behavior.	Forced variables remain in the final VIF set. **PASS I**
VAR-03	Render extreme VIF safely	Injected/inspected high and infinite VIF presentation cases.	High values warn without blocking; extreme/infinite values render explicitly. **PASS I**
VAR-04	Display coefficient context	Ran a slice and loaded Variables context.	Current coefficient is shown per variable. **PASS I**
VAR-05	Display period due-to context	Inspected Y1, Y2, and Total values from the last-run store.	Due-to values match stored model contributions. **PASS I**
VAR-06	Generalize to three model years	Ran a 156-week slice and inspected context periods.	Y1/Y2/Y3/Total options appear dynamically. **PASS I F**
VAR-07	Gate context by product	Switched the selected product after a run.	Context blanks until the new product is run. **PASS I**
VAR-08	Gate context by channel	Switched to another channel with the same product.	Same-channel showed nine variables; other-channel showed zero context rows. **PASS I**
VAR-09	Gate context by target	Switched target without rerunning.	Context blanks; edits remain available. **PASS I**
VAR-10	Preserve editor state when context blanks	Changed selection after editing the config table.	Context columns clear without discarding editor rows. **PASS I**
VAR-11	Auto-persist Variables Run	Changed adstock from 0.5 to 0.91 and clicked Run without Save.	Scoped config was written before fitting and the change was used. **PASS after fix I**
VAR-12	Keep batch run from overwriting editor	Executed the batch single-run path while editor data existed.	Batch path did not persist editor rows. **PASS I**

ID	Objective	Method / scenario	Result and evidence
VAR-13	Reject invalid equal bounds	Set lower bound equal to upper bound on Variables Run.	Run aborted visibly; old config was not silently used. **PASS after fix I R**
VAR-14	Hide excluded rows safely	Enabled the native excluded-row filter and then saved.	Display filtering did not drop excluded rows from saved data. **PASS I**
VAR-15	Remove superseded columns	Inspected the final Variables schema.	Temporary was-to-now columns were removed. **PASS I**

5.3 C. Two-Tier Configuration, Callback Ordering, and CLI Consistency

ID	Objective	Method / scenario	Result and evidence
CFG-01	Default config resolution	Resolved a slice with no channel override.	Product default path selected. **PASS I**
CFG-02	Override precedence	Created a channel override and resolved the same slice.	Override selected ahead of default. **PASS I**
CFG-03	Default filename convention	Generated the default path for dataset/product.	Path follows the documented dataset/product naming pattern. **PASS I**
CFG-04	Override filename convention	Generated a channel-specific path.	Path adds the normalized `__ch_<channel>` suffix. **PASS I**
CFG-05	Channel save isolation	Saved a channel override while hashing the default.	Override created; default remained byte-for-byte unchanged. **PASS I**
CFG-06	Editor scope status	Loaded default, inherited channel, and existing override states.	Status line correctly distinguishes all three. **PASS I**
CFG-07	Scoped Variables Run	Ran from default and channel scopes.	Each run used the exact file being edited (WYSIWYG). **PASS I**
CFG-08	Batch override semantics	Ran batch resolution with an existing override.	Batch used override-wins behavior. **PASS I**
CFG-09	Scoped regenerate	Regenerated in channel scope and hashed the default.	Only the override was regenerated. **PASS after fix I**

ID	Objective	Method / scenario	Result and evidence
`CFG-10`	Reset existing override	Clicked Reset to default with an existing channel file.	Override deleted; resolution fell back to default. **PASS after fix I**
`CFG-11`	Reset absent override	Clicked Reset when no override existed.	No destructive action or false revision bump occurred. **PASS I**
`CFG-12`	Prevent stale channel scope	Removed/cleared channel while channel scope was selected.	Scope forced back to Product default. **PASS after fix I**
`CFG-13`	Write-before-read ordering	Inspected callbacks after `config_rev` refactor.	Readers depend on the post-write signal, not raw button clicks. **PASS after fix I**
`CFG-14`	Save revision signal	Performed a successful Save and an invalid Save.	Revision bumps only after the successful write. **PASS I**
`CFG-15`	Regenerate revision signal	Performed a scoped regeneration.	Revision bumps only after the file update. **PASS I**
`CFG-16`	Reset revision signal	Deleted an override and repeated the no-op reset.	Real delete bumps; no-op returns `no_update`. **PASS I**
`CFG-17`	Variables Run revision signal	Persisted config and then simulated model failure.	Revision still bumped after persist, refreshing status despite fit failure. **PASS after fix I**
`CFG-18`	Invalid Variables Run revision	Passed invalid bounds through Variables Run.	No file write and no phantom revision bump. **PASS I**
`CFG-19`	Shared resolver in pipeline	Compared dashboard, batch, and CLI path results.	All delegated to the same pipeline resolver. **PASS after fix I**
`CFG-20`	Batch resolves per channel	Inspected and executed `run_all.py` resolution loop.	Each product-channel slice resolves independently. **PASS I**
`CFG-21`	Single-slice runner parity	Ran/inspected `run_brand1_channel1.py`.	Tiered config is preferred; no legacy root fallback remains. **PASS after fix I**
`CFG-22`	Config role is ACV authority	Set ACV role to auto and force in separate CLI-style runs.	Auto was not forced; force was included. **PASS after fix I**

ID	Objective	Method / scenario	Result and evidence
CFG-23	Remove hidden CLI forcing	Inspected `run_all.py` base config.	No unconditional ACV `force_include` remains. **PASS I**
CFG-24	Single default-creation entry	Deleted/restored a default under backup and reloaded through dashboard.	Dashboard and CLI both recreate through the shared pipeline helper. **PASS after fix I**
CFG-25	Lazy workbook read	Called default loading with an existing config.	Excel was not read on the common existing-file path. **PASS after fix I**
CFG-26	Unified mutation lock identity	Compared lock objects across pipeline and dashboard.	Default creation, save/run, regenerate, reset, and upload share one RLock. **PASS after fix I**
CFG-27	Concurrent in-process writes	Ran two channel uploads in separate threads.	Both completed without corruption or temp artifacts. **PASS I**
CFG-28	Multi-process limitation disclosure	Reviewed lock scope and documentation.	Process-local limitation is stated; OS locking is not overclaimed. **PASS I**

5.4 D. Variable Template Validation and Transactional Writes

ID	Objective	Method / scenario	Result and evidence
TPL-01	Canonical template schema	Downloaded and inspected the workbook columns.	Nine canonical fields matched the documented order. **PASS I**
TPL-02	Excel download serialization	Decoded the Dash download payload and opened it as XLSX.	Workbook was valid and editable. **PASS I**
TPL-03	CSV upload parsing	Uploaded a valid CSV template.	Rows parsed and entered the same validation path. **PASS I**
TPL-04	XLSX upload parsing	Uploaded a valid Excel template.	Rows parsed and entered the same validation path. **PASS I**

ID	Objective	Method / scenario	Result and evidence
`TPL-05`	Default routing	Uploaded a group with `channel=ALL`.	Product default targeted. **PASS I**
`TPL-06`	Override routing	Uploaded a group with a specific channel.	Channel override targeted. **PASS I**
`TPL-07`	Unlisted means excluded	Uploaded three listed variables for a 79-variable config.	Remaining 76 variables became excluded. **PASS I**
`TPL-08`	Preserve listed edits	Changed bounds/role/adstock in listed rows.	Uploaded values were preserved after round-trip. **PASS I**
`TPL-09`	Runtime override precedence	Uploaded default and channel groups then resolved the slice.	Channel override won at runtime. **PASS I**
`TPL-10`	Unknown product rejection	Uploaded a product not present in the workbook.	Upload rejected; no write occurred. **PASS after fix I**
`TPL-11`	Unknown channel rejection	Uploaded an invalid channel name.	No orphan override file was created. **PASS after fix I**
`TPL-12`	Unknown variable rejection	Introduced a variable-name typo.	Upload rejected rather than silently excluding the intended variable. **PASS after fix I**
`TPL-13`	Duplicate row rejection	Repeated the same product/channel/variable row.	Upload rejected before writes. **PASS after fix I**
`TPL-14`	Bound ordering validation	Set lower bound greater than or equal to upper bound.	Upload rejected with a clear validation result. **PASS I**
`TPL-15`	Multi-group input atomicity	Combined one valid and one invalid group.	Neither group was written. **PASS after fix I**
`TPL-16`	Rejected upload revision behavior	Passed a fully invalid file through the callback.	Store/revision remained unchanged. **PASS after fix I**
`TPL-17`	Case normalization before grouping	Mixed `ALL`/`all` and channel letter case.	Equivalent groups collapsed to one canonical target. **PASS after fix I**
`TPL-18`	Whitespace normalization	Uploaded padded product/channel/variable values.	Values normalized before validation/routing. **PASS I**
`TPL-19`	Empty/no-op upload	Uploaded headers with no valid rows.	Upload rejected and no revision bump occurred. **PASS after fix I**

ID	Objective	Method / scenario	Result and evidence
`TPL-20`	Unique stage files	Inspected staged filenames for a multi-target upload.	UUID-based temp names prevent same-process collisions. **PASS after fix I**
`TPL-21`	Serialization failure cleanup	Injected a non-I/O serialization exception.	Exception caught; partial temp removed; no target written. **PASS after fix I**
`TPL-22`	Commit failure with existing targets	Injected failure on the second replace after backing up two targets.	First target restored; second stayed intact. **PASS after fix I**
`TPL-23`	Commit failure with new targets	Injected failure after creating the first new target.	New target was removed during rollback. **PASS after fix I**
`TPL-24`	Rollback restore failure	Injected an error while restoring an original.	Status became ROLLBACK INCOMPLETE and the last backup copy was preserved. **PASS after fix I**
`TPL-25`	Rollback report accuracy	Used three slices: one restored, one restore-failed, one uncommitted.	Only the genuinely changed slice was reported as changed. **PASS after fix I**
`TPL-26`	Three-state callback behavior	Exercised applied, rejected, and rollback-incomplete results.	Applied/partial refresh UI; rejected leaves state untouched. **PASS after fix I**
`TPL-27`	Happy-path multi-target commit	Uploaded two valid channel groups.	Both targets committed and backups were removed. **PASS I**
`TPL-28`	Post-test cleanup	Scanned configs for `uploadtmp`, `.bak`, and test overrides.	No test artifacts remained. **PASS I F**

5.5 E. Export Tab and Definitions

ID	Objective	Method / scenario	Result and evidence
`EXP-01`	Register Export tab	Loaded navigation and switched screens.	Export screen and controls rendered; callbacks registered. **PASS I**
`EXP-02`	Single-slice workbook	Exported Brand 1 x Channel 1.	Download payload was a valid XLSX. **PASS I**

ID	Objective	Method / scenario	Result and evidence
EXP-03	Fit statistics sheet	Opened `fit_statistics`.	R-squared, adjusted R-squared, MAPE, holdout, grade, and counts were present. **PASS I**
EXP-04	Fit structure sheet	Opened `fit_structure`.	Columns exactly match the upload template, enabling edit/re-upload. **PASS I**
EXP-05	Weekly due-tos sheet	Opened `weekly_due_tos`.	Actual dates and tidy driver contributions were present. **PASS I**
EXP-06	Full reported date coverage	Compared weekly due-to dates against the reported fit window.	Zero missing dates after correcting the old split-only slice. **PASS after fix I**
EXP-07	All-slice aggregation	Ran All combinations on a bounded one-product subset.	Nine runnable slices aggregated successfully. **PASS I**
EXP-08	Fresh-data default creation	Removed/restored a default around export.	Export created the default through the shared entry and succeeded. **PASS after fix I**
EXP-09	Missing target isolation	Requested a target absent from one product in All mode.	Slice went to errors sheet; enumeration did not raise KeyError. **PASS after fix I**
EXP-10	Grade formatting in Excel	Inspected openpyxl styles on the grade column.	Moderate received amber fill and readable bold text; other grades follow their colors. **PASS after fix I**
EXP-11	All-slices-failed delivery	Forced every slice to fail.	Workbook still downloaded with accessible errors sheet. **PASS after fix I**
EXP-12	Input guards	Called export without required target/window/slice selections.	Clear status returned; no invalid download attempted. **PASS I**
DEF-01	Register Definitions tab	Loaded navigation and rendered the new screen.	All sections displayed without missing IDs. **PASS I**
DEF-02	Average weekly due-to definition	Compared text to engine aggregation logic.	Media uses execution weeks; other drivers use period weeks. **PASS I**

ID	Objective	Method / scenario	Result and evidence
DEF-03	Grade range precision	Checked threshold wording and code constants.	Moderate is explicitly greater than 10% and at most 40%. **PASS after fix I**
DEF-04	Adstock scope precision	Compared text to variable specs.	Adstock applies to any variable with decay, typically but not exclusively media. **PASS after fix I**
DEF-05	Config precedence precision	Compared text to Variables and batch behavior.	Batch/export/CLI override-wins; Variables Run is WYSIWYG. **PASS after fix I**
DEF-06	Holdout formula revision	Updated definitions after reserved-tail engine change.	Text now reflects the always-reserved validation tail, with short-history/zero exceptions. **PASS after fix I**
DEF-07	Method structure wording	Checked "same coefficients" versus "same structure/process".	Definitions correctly state validation selects structure and full-window refit changes coefficients. **PASS after fix I**
DEF-08	Threshold interpolation	Changed/read constants through definition rendering.	Displayed thresholds are derived from code, not stale literals. **PASS I**

5.6 F. Modeling Engine, Holdout, and Three-Year Robustness

ID	Objective	Method / scenario	Result and evidence
MOD-01	Always reserve validation tail	Ran 52-, 104-, and 156-week windows on 156-week history.	Each normal run retained an out-of-sample tail of up to 13 weeks. **PASS I F**
MOD-02	Avoid zero-holdout at full history	Ran 156-week window on 156 weeks.	Holdout MAPE was real, not NaN. **PASS after fix I F**
MOD-03	Validation structure selection	Compared validation-selected variables with reported coefficient variables.	Sets matched across tested brands/windows. **PASS after fix I**

ID	Objective	Method / scenario	Result and evidence
`MOD-04`	No validation-tail leakage	Inspected selection data boundaries.	Structure selection used only pre-tail data. **PASS I**
`MOD-05`	Full-window reported refit	Checked reported fit/contribution row counts.	Coefficients and due-tos use the complete selected window. **PASS I**
`MOD-06`	Preserve holdout metric	Compared in-sample MAPE and validation-tail MAPE.	Metrics remain distinct; in-sample fit is not relabeled holdout. **PASS I**
`MOD-07`	104-week backward consistency	Compared revised validation holdout with the prior training model.	Holdout reproduced approximately 31.83%. **PASS I F**
`MOD-08`	156-week holdout behavior	Ran Brand 1 x Channel 1 at 156 weeks.	Final holdout MAPE approximately 58.31%. **PASS I F**
`MOD-09`	52-week final regression	Ran Brand 1 x Channel 1 at 52 weeks.	R2 0.946497; holdout MAPE 19.3178%; nine predictors. **PASS F**
`MOD-10`	104-week final regression	Ran Brand 1 x Channel 1 at 104 weeks.	R2 0.883345; holdout MAPE 31.8339%; nine predictors. **PASS F**
`MOD-11`	156-week final regression	Ran Brand 1 x Channel 1 at 156 weeks.	R2 0.810031; holdout MAPE 58.3078%; nine predictors. **PASS F**
`MOD-12`	Multi-brand smoke	Ran at least one additional product/slice after engine changes.	Fit completed without structure/refit mismatch. **PASS I**
`MOD-13`	Three model-year contribution table	Inspected 156-week `contrib_by_year`.	Three year columns plus dynamic YoY fields rendered. **PASS I F**
`MOD-14`	Diagnostics fitted-line length	Compared fitted values with full reported dates.	Fitted line spans all reported weeks. **PASS after fix I**
`MOD-15`	CLI fitted-line length	Checked single-slice runner plotting after refit.	CLI plot uses full-window fitted values and tail forecast. **PASS after fix I**
`MOD-16`	CLI week labels	Inspected result field names after refit.	Uses `n_weeks_reported` and `holdout_weeks`; no misleading split label. **PASS after fix I**

ID	Objective	Method / scenario	Result and evidence
MOD-17	Sponsor estimator equivalence	Ran the equivalence script from the repository root.	PASS; final observed scale-normalized gap 1.69×10^{-8}. **PASS R F**
MOD-18	Scale-robust equivalence metric	Tested a coefficient near zero.	Metric no longer explodes by dividing by an approximately zero coefficient. **PASS after fix I**
MOD-19	Equivalence script path portability	Ran the script outside the `code` directory.	Data path resolves relative to the script; root invocation succeeds. **PASS after fix R F**
MOD-20	Current model-year counts	Read final regression stores for all windows.	52/104/156 produce 1/2/3 model years respectively. **PASS F**

5.7 G. Contributions Time Selection, Mapping Upload, and Comparison

ID	Objective	Method / scenario	Result and evidence
TIME-01	Place period filter correctly	Inspected Contributions card header.	Filter is next to Avg weekly due-to per driver. **PASS I**
TIME-02	Entire-window equivalence	Recomputed period aggregation and compared with engine average.	Maximum difference was on the order of 10^{-11}. **PASS I**
TIME-03	Media execution-week mask	Compared selected weeks with raw execution indicators.	Media averages only positive-execution weeks within the selected period. **PASS I**
TIME-04	Non-media period average	Compared controls against all selected weeks.	Non-media averages use every week in the period. **PASS I**
TIME-05	Year-period counts	Checked Y1/Y2/Y3 selections on 156-week data.	Each model year contains 52 weeks. **PASS I**
TIME-06	Latest 13 weeks	Resolved the rolling selection.	Exactly the latest 13 rows selected. **PASS I**
TIME-07	Latest 4 weeks	Resolved the rolling selection.	Exactly the latest four rows selected. **PASS I**

ID	Objective	Method / scenario	Result and evidence
`TIME-08`	Calendar Q4	Resolved latest October–December weeks.	Q4 2025 selected 13 weeks, 2025-10-05 through 2025-12-28. **PASS after fix R F**
`TIME-09`	Stable Current year token	Compared 104- and 156-week runs.	Both map Current to 2025-01-05 through 2025-12-28. **PASS after fix R**
`TIME-10`	Stable Previous year token	Compared 104- and 156-week runs.	Both map Previous to 2024-01-07 through 2024-12-29. **PASS after fix R**
`TIME-11`	One-year Previous option	Ran a 52-week window.	Previous is hidden and resolves to an empty index defensively. **PASS after fix R**
`TIME-12`	Clamp stale Previous	Passed `previous` into a new 52-week run.	Initial chart/caption rendered Entire window, never a mislabeled Previous. **PASS after fix R**
`TIME-13`	Clear store on missing input	Called Run without a datapath.	Charts blanked and `contrib_wk=None`. **PASS after fix R**
`TIME-14`	Clear store on invalid bounds	Emulated Variables trigger with equal bounds.	Run aborted and contribution store cleared. **PASS R**
`TIME-15`	Clear store on model exception	Injected a `run_slice` exception.	Failure tuple cleared the contribution store. **PASS R**
`TIME-16`	Catch bad data path	Passed a truthy nonexistent workbook path.	Callback returned Model failed and cleared the store; no exception escaped. **PASS after fix R**
`TIME-17`	Empty store toggle	Called period redraw with no contribution store.	Returned `no_update` and did not repaint stale data. **PASS R**
`TIME-18`	Time-map valid CSV	Uploaded date/period CSV.	Store created period-to-date lists. **PASS I R**
`TIME-19`	Time-map valid XLSX	Uploaded Excel dates with labels.	Workbook parsed through the same map builder. **PASS R**
`TIME-20`	Multiple periods per date	Mapped one week to Current Year and My Q4.	Both mappings retained. **PASS R**

ID	Objective	Method / scenario	Result and evidence
`TIME-21`	Deduplicate repeated rows	Repeated an identical date/period row.	Stored date appears once. **PASS R**
`TIME-22`	Run-scoped mapped options	Included present and absent mapped periods.	Present period shown; absent period hidden. **PASS I R**
`TIME-23`	Mapped Q4 selection	Mapped the 13 Q4 weeks to My Q4 and selected it.	Exactly 13 weeks used; caption includes "mapped". **PASS I R**
`TIME-24`	Missing-column map	Uploaded a file without required columns.	Clear error; existing store untouched. **PASS R**
`TIME-25`	Mixed valid/invalid dates	Uploaded valid, empty, and garbage dates together.	Bad rows skipped and reported by spreadsheet row number; valid rows loaded. **PASS after fix R**
`TIME-26`	All-invalid dates	Uploaded only empty/garbage dates.	Clear message and `no_update` store result. **PASS after fix R**
`TIME-27`	Mixed date formats	Mixed ISO and MM/DD/YYYY strings in one CSV.	Both valid dates parsed with mixed-format handling. **PASS after fix R**
`TIME-28`	Same-name map refresh	Re-uploaded map X with a different assigned week.	Because `time_map` is an Input, chart value updated from 1 to 9. **PASS after fix R**
`TIME-29`	Removed mapped period clamp	Removed the selected mapped label.	Redraw clamps to Entire window. **PASS after fix R**
`TIME-30`	No-comparison fallback	Selected No comparison.	Single trace and single-period caption retained. **PASS R**
`TIME-31`	Current vs Previous comparison	Ran 104 weeks with Current vs Previous.	Two grouped traces share driver order; caption reports 52 vs 52 weeks. **PASS R**
`TIME-32`	Invalid comparison clamp	Passed a missing mapped compare token.	Single trace returned. **PASS R**
`TIME-33`	Exclude primary from compare options	Built compare options with Current selected.	Current token absent from compare list. **PASS after fix R**

ID	Objective	Method / scenario	Result and evidence
`TIME-34`	Reset collision after primary change	Changed primary to the current compare token.	Compare callback resets value to none. **PASS after fix R**
`TIME-35`	Defensive A-vs-A guard in redraw	Called redraw with identical period and compare.	Single trace returned; no "A vs A" caption. **PASS after fix R**
`TIME-36`	Defensive A-vs-A guard in Run	Ran the model with identical period/compare state.	Run output contained one trace. **PASS after fix R**
`TIME-37`	Same dates, distinct tokens	Compared built-in Q4 with mapped MyQ4 over the same 13 dates.	Two traces remained allowed, preserving a mapping-validation use case. **PASS R**
`TIME-38`	Map re-upload callback dependency	Inspected callback map after E3 fix.	Redraw inputs include period, compare, and time map; contribution store remains State. **PASS R**
`TIME-39`	Compare option dependency	Inspected callback map after A-vs-A fix.	Compare options depend on primary period and reset collisions. **PASS R**
`TIME-40`	Final 52/104/156 trace behavior	Ran Current vs Previous state across all windows.	52 clamps to one trace; 104 and 156 render two traces. **PASS F**

5.8 H. Final Compilation, Registration, and Workspace Hygiene

ID	Objective	Method / scenario	Result and evidence
`SMK-01`	Compile pipeline	Ran `python -B -m py_compile` on `capstone_pipeline.py`.	No syntax/import compile error. **PASS F**
`SMK-02`	Compile dashboard	Ran `python -B -m py_compile` on `dashboard.py`.	No syntax/import compile error. **PASS F**
`SMK-03`	Compile batch runner	Compiled `run_all.py`.	No syntax/import compile error. **PASS F**
`SMK-04`	Compile single-slice runner	Compiled `run_brand1_channel1.py`.	No syntax/import compile error. **PASS F**
`SMK-05`	Compile equivalence script	Compiled the sponsor-equivalence script.	No syntax/import compile error. **PASS F**
`SMK-06`	Register final dashboard	Imported dashboard and ran the custom wiring validator.	66 components / 27 callbacks; no missing component IDs. **PASS F**

ID	Objective	Method / scenario	Result and evidence
SMK-07	Config diff hygiene	Ran `git diff - configs`.	No config differences. **PASS F**
SMK-08	Temp/backup hygiene	Scanned configs for upload temps, backup files, and test channel overrides.	No matching artifacts. **PASS F**
SMK-09	Preserve unrelated batch output	Reviewed Git status after final regression.	Pre-existing modified batch summary left untouched. **PASS R F**
SMK-10	Source-count precision	Compared final tracked source list with the report/tracker.	Five source files plus one tracker document, not six source files. **PASS R**
SMK-11	Final equivalence from root	Invoked the sponsor comparison from repository root.	Path-independent execution and PASS result. **PASS F**
SMK-12	Final PowerPoint closure	Compared the final tracker to all groups and repeated edge fixes.	Every PowerPoint item is complete in the final tested state. **PASS R F**

6 Defect Discovery and Retest Summary

The session did not stop at happy-path verification. Review findings were converted into explicit regression tests. The most consequential defect families were:

1. **Editor-to-run correctness:** Variables Run originally did not persist on-screen edits. The fix was retested with a concrete adstock change and with invalid bounds.
2. **Slice context correctness:** Context originally checked only product. Channel and target gating were added and tested.
3. **Configuration scope correctness:** Regenerate, Reset, no-channel fallback, and CLI resolution each had separate divergence or scope defects that were closed.
4. **Callback ordering:** Independent Save/Reset/read callbacks created races. A post-write `config_rev` dependency was introduced and all writers were verified.
5. **Config authority:** Hidden CLI ACV forcing and the legacy root fallback could contradict the app. Both were removed and role-based behavior was tested in both directions.
6. **Template failure safety:** Unknown names, duplicates, bad channels, case collisions, empty uploads, partial validation, staging failures, mid-commit failures, restore failures, and concurrency were all fault-injected.
7. **Export failure safety:** New datasets, missing targets, colored grades, and all-failed workbooks were tested explicitly.
8. **Modeling validity:** The first full-refit implementation allowed validation and reported models to select different structures. The final design selects structure without tail leakage and refits that same structure on full data.

9. **Time-state correctness:** Contribution state clearing, 52-week Previous behavior, missing paths, map replacement, malformed dates, mixed formats, and A-vs-A comparison were each found during review and retested after repair.

7 Final Regression Snapshot

Window	R2	Adj. R2	MAPE	Holdout	Selected	Years
52 weeks	0.946497	0.935032	7.8611%	19.3178%	9	1
104 weeks	0.883345	0.872175	9.6023%	31.8339%	9	2
156 weeks	0.810031	0.798320	13.1984%	58.3078%	9	3

These values are a reproducibility snapshot for Brand 1 x Channel 1 with target Volume Sales on the local dataset and current config state. They are not universal performance guarantees for other products, channels, targets, configurations, or future data.

8 Reproducible Commands

```
# Compile the five modified Python sources
python -B -m py_compile code/capstone_pipeline.py code/dashboard.py \
  code/reference_alex_curvefit.py code/run_all.py \
  code/run_brand1_channel1.py

# Validate the final Dash component/callback graph
python -B -c "import sys; sys.path.insert(0,'code'); \
import dashboard; print(dashboard._validate_wiring())"

# Run the sponsor-estimator equivalence test from the repository root
python -B code/reference_alex_curvefit.py

# Confirm no config drift or transaction artifacts
git diff -- configs
Get-ChildItem configs -File | Where-Object {
  $_.Name -match 'uploadtmp|\.bak$|__ch_'
}
```

9 Limitations and Remaining Non-PowerPoint Work

- The runtime/scaling benchmark and its email are not included in the completed PowerPoint batch and remain a separate action unless completed elsewhere.
- The grade thresholds (Good at most 10%, Moderate above 10% and at most 40%, Bad above 40%) are an internal product rule rather than a threshold specified in Alex's slides. The implementation is tested, but sponsor confirmation remains a policy question.
- The configuration write lock is process-local. Multi-process deployments would require an OS-level or external distributed lock.
- The current numeric regression values are dataset/config specific. Functional expectations, not exact metrics, should be used for future regression data.

10 Conclusion

The final code state passed the named functional, failure-safety, modeling, callback, export, and regression tests recorded in this report. The review process materially strengthened the implementation: several initially plausible “done” states were downgraded, corrected, and only then retested to closure. On the evidence available in this workspace and session, every item from Alex’s annotated PowerPoint is implemented and verified.

References

[1] Alex Hathcock. *Dashboard Edits*. Six-slide annotated PowerPoint supplied to the Big Chalk Capstone team, July 2026.

[2] Big Chalk Capstone Team. *Alex's Dashboard Edits – Plan & Tracker*. `docs/Alex_Dashboard_Edits_Plan.md`, accessed July 23, 2026.

[3] Big Chalk Capstone Team. Dashboard implementation and callback graph. `code/dashboard.py`, working-tree version reviewed July 23, 2026.

[4] Big Chalk Capstone Team. Marketing-mix modeling pipeline. `code/capstone_pipeline.py`, working-tree version reviewed July 23, 2026.

[5] Big Chalk Capstone Team. Batch and single-slice runners. `code/run_all.py` and `code/run_brand1_channel1.py`, working-tree versions reviewed July 23, 2026.

[6] Big Chalk Capstone Team. Sponsor-estimator equivalence test. `code/reference_alex_curvefit.py`, working-tree version reviewed July 23, 2026.

[7] Big Chalk Capstone Project. *Anonymized Data for Project.xlsx*. Local test dataset, SHA-256 beginning `C4303D04B809A805`, inspected July 23, 2026.

[8] Boqi Niu and Codex. Implementation and independent review session for Alex's dashboard edits. Session evidence through July 23, 2026.